Things we know:

- $B = B_m + B_{m-1} + B_{m-2} + \dots$ grouping by # of boundary propagators

↳ X_{1j} and X_{2j} for all j

(using the $c_{14} = c_{15} = \dots = c_{1n-1} = 0$ 1-zero)

if B satisfies 1-zero so does each B_i

send $X_{1j} \to \lambda X_{1j}$, $X_{2j} \to \lambda X_{2j}$ then $B_i \to \lambda^i B_i$

ex: $B = \underbrace{\frac{1}{X_{13}X_{24}}}_{B_{-2}} + \underbrace{\frac{1}{X_{13}X_{35}}}_{B_{-1}} \to \frac{1}{\lambda^2 X_{13}X_{24}} + \frac{1}{\lambda X_{13}X_{35}}$

Prop: if $B=0$ when the X_{ij} satisfy each constraint, then the coefficients in B are all equal.

② We can split each B_i by D-subsets, and each D-subset must be zero.

let's split variables: x represents all boundary propagators, y all other ones

$B = B(x,y) = \sum_i B_i(x,y)$ constraint looks like $L(x)=0$ ← linear

multi-index notation: $B_i(x,y) = \sum_{\alpha,\beta} \frac{a_{\alpha\beta}}{x^\alpha y^\beta}$ → multi-index

- suppose that $L(x)=0 \Rightarrow B_i(x,y)=0$

since L does not depend on y, for each fixed β the term → represent a combination of powers of y-terms

$$\boxed{\sum_\alpha \frac{a_{\alpha\beta}}{y^\beta x^\alpha} = 0} \quad (\forall x \text{ s.t. } L(x)=0)$$

→ this is a D-subset

other way of thinking:

$$B = \sum_{\substack{\alpha,\beta \\ |\alpha|+|\beta|=n-3}} \frac{a_{\alpha\beta}}{x^\alpha y^\beta} = \sum_i \underbrace{\sum_{\substack{\alpha,\beta \\ |\alpha|+|\beta|=n-3 \\ |\alpha|=i}} \frac{a_{\alpha\beta}}{x^\alpha y^\beta}}_{=B_i}$$

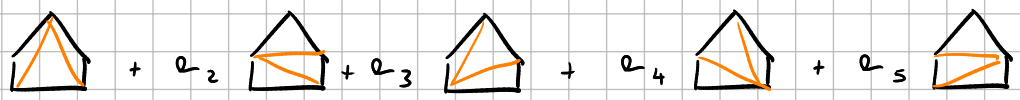
$$B_i = \sum_{\substack{\alpha,\beta \\ |\alpha|=i}}$$

$\alpha_1+\alpha_2+\alpha_3+\dots$

$$= \sum_i \sum_{\substack{\beta \\ |\beta|+i=n-3}} \underbrace{\sum_{\substack{\alpha \\ |\alpha|=i}} \frac{a_{\alpha\beta}}{x^\alpha y^\beta}}_{\text{D-subset}}$$

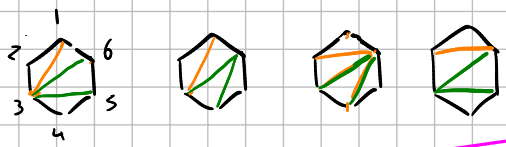
---

ex: $n=5$

$B =$ 

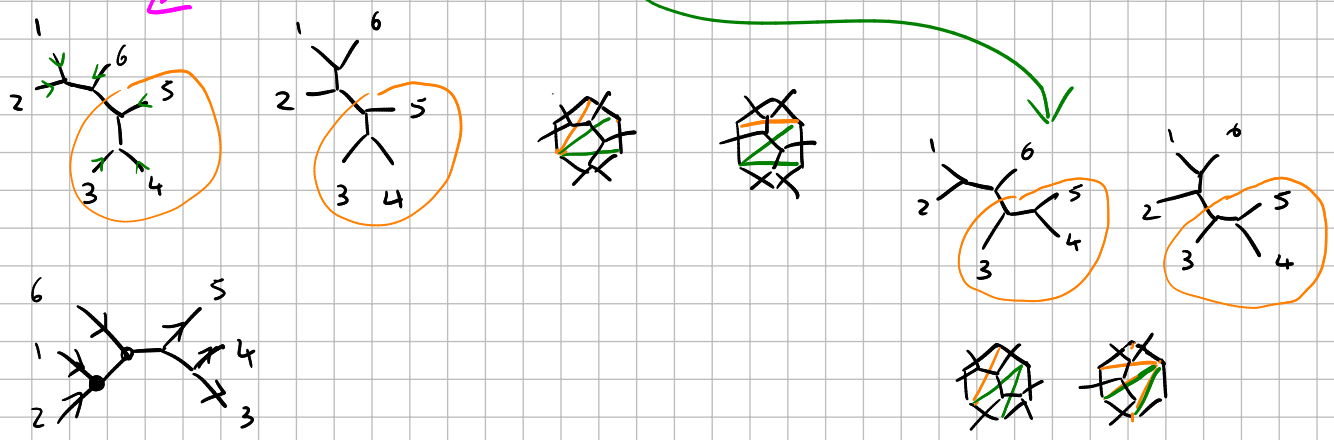
$$= \underbrace{\frac{a_1}{x_{13}x_{14}} + \frac{a_2}{x_{24}x_{25}}}_{B_{-2}} + \underbrace{\frac{a_3}{x_{35}x_{31}} + \frac{a_4}{x_{42}x_{41}} + \frac{a_5}{x_{52}x_{53}}}_{B_{-1}}$$

---

ex $\boxed{n=6}$ $B_{-1} =$ 

D-subset

$$= \frac{a_1}{x_{13}x_{36}x_{35}} + \underbrace{\frac{a_2}{x_{13}x_{36}x_{46}} + \frac{a_3}{x_{26}x_{36}x_{46}}}_{\text{D subset}} + \frac{a_4}{x_{26}x_{35}x_{36}}$$

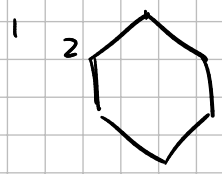


Now want to prove that each D-subset satisfies the zero only if all the coefficients in it are equal.

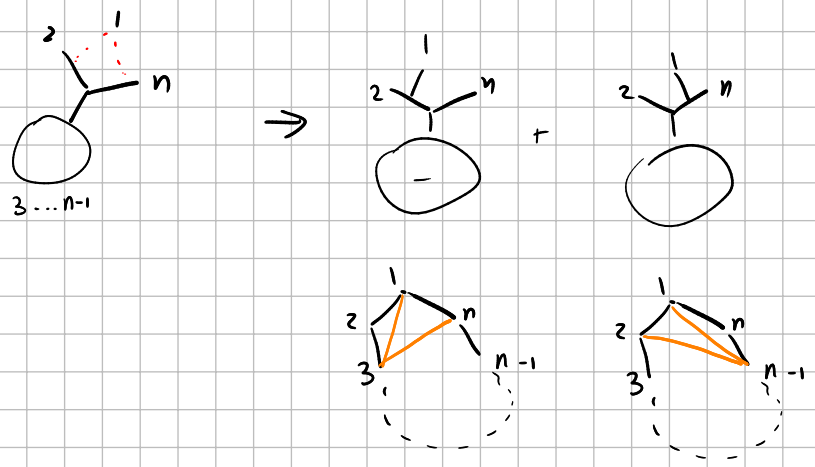
To obtain a D-subset, we pick a diagram, remove leg 1, then reattach it in all possible ways (compatible with ordering)

---

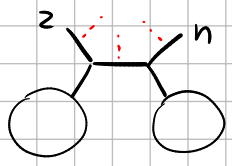
0 boundary prop: not allowed (non-bad)



1 boundary prop:

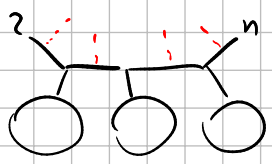


2 boundary prop

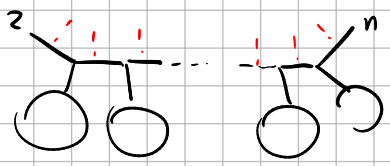


3 terms

3 boundary prop

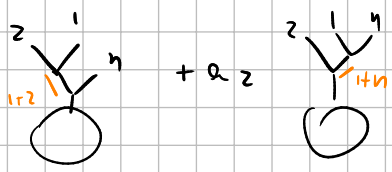


k boundary prop



k+1 terms

$k=1$

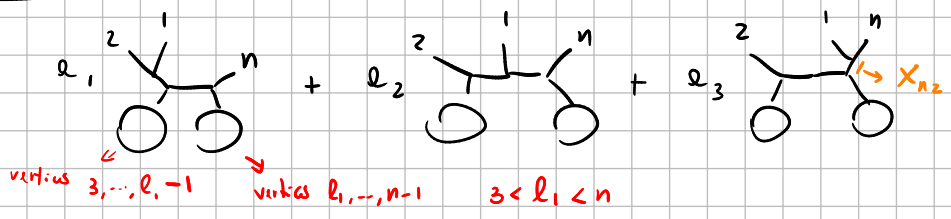


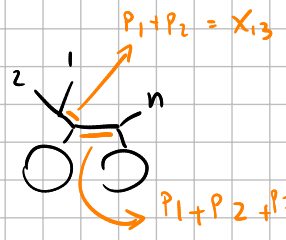
$$\propto \frac{a_1}{s_{12}} + \frac{a_2}{s_{1n}} = \frac{a_1}{X_{13}} + \frac{a_2}{X_{2n}} = 0$$

by example when $L(x)=0$

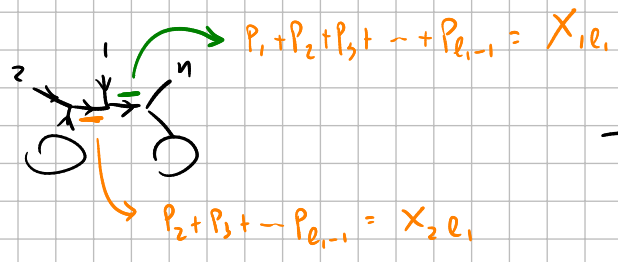
constraint says that $X_{2n} = -X_{13}$ so $a_1 = a_2$

$k=2$

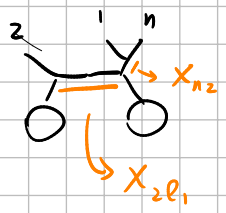




$$\longrightarrow \frac{1}{X_{13}\, X_{1\ell_1}}$$



$$\rightarrow \frac{1}{X_{1\ell_1}\, X_{2\ell_1}}$$



$$\rightarrow \frac{1}{X_{2\ell_1}\, X_{2n}}$$

D subset $\propto$ $\dfrac{a_1}{X_{13}\, X_{1\ell_1}} + \dfrac{a_2}{X_{1\ell_1}\, X_{2\ell_1}} + \dfrac{a_3}{X_{2\ell_1}\, X_{2n}} = 0$

numerator must be zero

<u>common denominator</u> $\propto$ $a_1 X_{2\ell_1} X_{2n} + a_2 X_{13} X_{2n} + a_3 X_{1\ell_1} X_{13} = 0$

in particular must be true in the limit $\begin{cases} X_{1\ell_1} \to 0 \\ X_{2\ell_1} \to -X_{13} \end{cases}$ "the $X_{1\ell_1}$ cut" (Rodine) $(X_{2\ell_1} = X_{1\ell_1} - X_{13})$ for all X_{1j}, X_{2j} satisfying the constraint

So $\propto$ $-a_1 X_{13} X_{2n} + a_2 X_{13} X_{2n} = 0$

$\rightarrow$ so $a_1 = a_2$

---

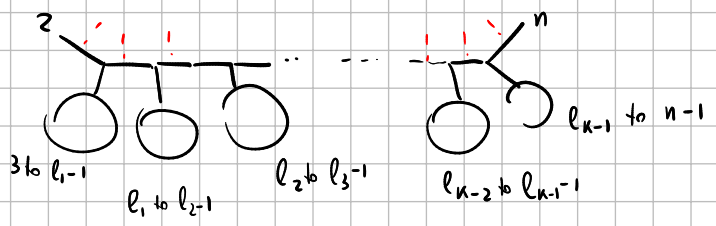
also true if $\begin{cases} X_{2\ell_1} \to 0 \\ X_{1\ell_1} \to X_{13} \end{cases}$ $\rightarrow 0 = a_2 X_{13} X_{2n} + a_3 X_{1\ell_1} X_{13} =$

$= -a_2 X_{13} X_{13} + a_3 X_{13} X_{13} \rightarrow a_2 = a_3$

$X_{2n} = -X_{13}$

So $a_1 = a_2 = a_3$

<u>now arbitrary K</u>

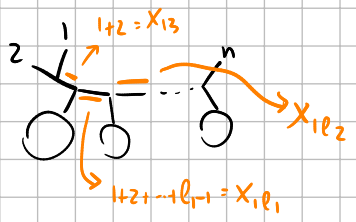


$l_0 = 3 \qquad l_k = n$

$$l_0 < \underbrace{l_1 < l_2 < \cdots < l_{k-1}}_{\text{arbitrary}} < l_k$$

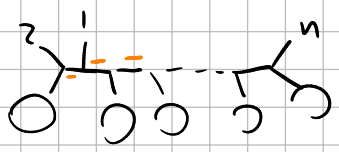
for triangulation

terms



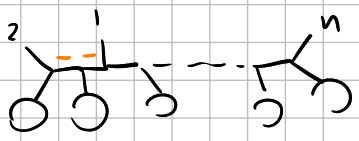
$$\frac{1}{X_{13}\, X_{1l_1} X_{1l_2} \cdots X_{1l_{k-1}}} = \frac{1}{\underbrace{X_{1l_0} X_{1l_1} \cdots X_{1l_{k-1}}}_{k \text{ terms}}}$$

next



$$\frac{1}{X_{2l_1}\, X_{1l_1}\, X_{1l_2} \cdots X_{1l_{k-1}}}$$

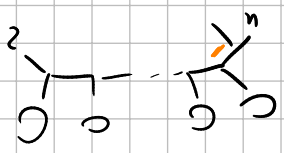
next



$$\frac{1}{X_{2l_1}\, X_{2l_2}\, X_{1l_2} \cdots X_{1l_{k-1}}}$$

⋮

at the end



$$\frac{1}{X_{2l_1}\, X_{2l_2} \cdots X_{2n}} = \frac{1}{X_{2l_1}\, X_{2l_2} \cdots X_{2l_k}}$$

D subset is a sum of products

$$\rightarrow \quad \sum_{i=0}^{k} \alpha_i \underbrace{\left(\prod_{b=1}^{i} \frac{1}{X_{2l_b}} \right)}_{i \text{ things}} \underbrace{\left(\prod_{a=i}^{k-1} \frac{1}{X_{1l_a}} \right)}_{k-i \text{ terms}}$$

generic D subset with k boundary propagators

k in total

common denominator

$$\frac{1}{\left(\prod_{b=1}^{k} X_{2l_b} \right) \left(\prod_{a=0}^{k-1} X_{1l_a} \right)}$$

numerator:

$$S = \sum_{i=0}^{k} \alpha_i \underbrace{\left(\prod_{a=0}^{i-1} X_{1l_a} \right) \left(\prod_{b=i+1}^{k} X_{2l_b} \right)}_{X_{1l_0} X_{1l_1} \cdots X_{1l_{i-1}} \; X_{2l_{i+1}} X_{2l_{i+2}} \cdots X_{2l_k}} = 0$$

note: • if $X_{1\ell_i} \to 0$ then $X_{2\ell_i} \to -X_{13} = X_{2n}$ $\quad (i = 1, \dots, k-1)$

• if $X_{2\ell_i} \to 0$ then $X_{1\ell_i} \to X_{13} = -X_{2n}$

---

① take $X_{1\ell_1} \to 0$ only 2 terms survive: $i=0, i=1$

is common so factors out

$$0 = S = \alpha_0 \underbrace{\left(\prod_{a=0}^{-1} X_{1\ell_a}\right)}_{=1} \left(\prod_{b=1}^{k} X_{2\ell_b}\right) + \alpha_1 \left(\prod_{a=0}^{0} X_{1\ell_a}\right) \left(\prod_{b=2}^{k} X_{2\ell_b}\right)$$

$$\propto \alpha_0 X_{2\ell_1} + \alpha_1 X_{1\ell_0} = -\alpha_0 X_{13} + \alpha_1 X_{13} \quad \to \quad \alpha_0 = \alpha_1$$

---

② take $X_{2\ell_{k-1}} \to 0$ only $i = k, k-1$ survive

common

$$0 = S = \alpha_k \left(\prod_{a=0}^{k-1} X_{1\ell_a}\right) \underbrace{\left(\prod_{b=k+1}^{k} X_{2\ell_b}\right)}_{=1} + \alpha_{k-1} \left(\prod_{a=0}^{k-2} X_{1\ell_a}\right) \left(\prod_{b=k}^{k} X_{2\ell_b}\right)$$

$$\propto \alpha_k \underbrace{X_{1\ell_{k-1}}}_{X_{13}} + \alpha_{k-1} \underbrace{X_{2\ell_k}}_{=n}$$

$$= \alpha_k X_{13} + \alpha_{k-1} X_{2n} = X_{13}(\alpha_k - \alpha_{k-1}) \quad \to \quad \alpha_k = \alpha_{k-1}$$

---

let $j \in \{1, \dots, k-2\}$

set $\begin{cases} X_{1\ell_{j+1}} \to 0 \\ X_{2\ell_j} \to 0 \end{cases}$ at some time

$$S = \sum_{i=0}^{k} \alpha_i \left(\prod_{a=0}^{i-1} X_{1\ell_a}\right) \left(\prod_{b=i+1}^{k} X_{2\ell_b}\right) = 0$$

if $i-1 \geq j+1$ 1st product vanishes

if $i+1 \leq j$ 2nd product vanishes

$\to$ need $\begin{cases} i-1 < j+1 \\ i+1 > j \end{cases} \to \begin{cases} i < j+2 \\ i > j-1 \end{cases} \to \begin{cases} i \leq j+1 \\ i \geq j \end{cases}$

only $i = j, j+1$ survive

common

$$0 = S = \alpha_j \left(\prod_{a=0}^{j-1} X_{1\ell_a}\right) \left(\prod_{b=j+1}^{k} X_{2\ell_b}\right) + \alpha_{j+1} \left(\prod_{a=0}^{j} X_{1\ell_a}\right) \left(\prod_{b=j+2}^{k} X_{2\ell_b}\right)$$

$$\propto \alpha_j \underbrace{X_{2\ell_{j+1}}}_{-X_{13}} + \alpha_{j+1} \underbrace{X_{1\ell_j}}_{X_{13}} = X_{13}(-\alpha_j + \alpha_{j+1}) \quad \to \quad \alpha_j = \alpha_{j+1}$$

So $\boxed{\alpha_0 = \alpha_1 = \alpha_2 = \cdots = \alpha_k}$

---

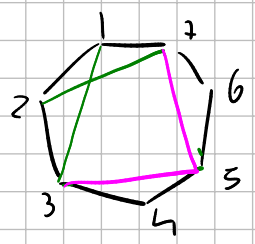
inline for non-local

D subset with 1 boundary projector:

$$\frac{1}{(\ - - -\ )} \left(\sum_{i<j} \frac{a_{ij\beta}}{x_{ij}} \right)$$

$a_{13\beta} = a_{2n\beta}$

all others are zero

β skipping what's here

strictly product of $n-4$ free chords (the triangle ones $x_{35}\, x_{36} \cdots$)